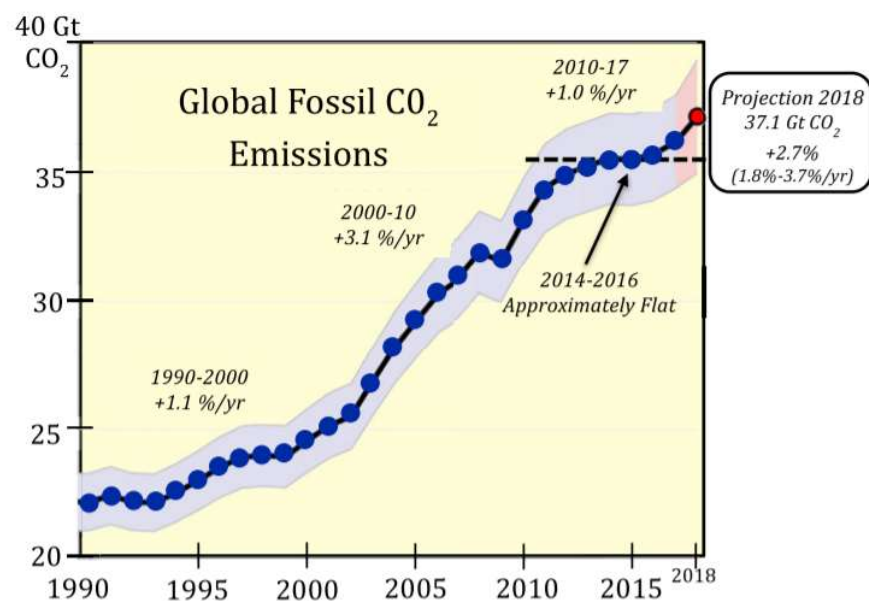


Scenario of Global Carbon Emissions

Population growth
Industrialization



Heavy dependence on fossil fuels



Our modern technological society, coupled with largely uncontrolled population growth, places extreme stress on the environment

How much CO₂ is being produced globally

Composition of gases in the Earth's atmosphere (volume in %)

Nitrogen ~78%

Oxygen ~ 21%

Carbon dioxide ~ 0.04%

The radius of Earth ~ 6371 km

Volume of Earth ~ $1.083 \times 10^{21} \text{ m}^3$

Volume of Troposphere ~ $6.13 \times 10^{18} \text{ m}^3$

~80% of mass of Earth's atmosphere is in the troposphere



1

$$\begin{aligned}\text{Volume of CO}_2 \text{ in the troposphere} &= 6.13 \times 10^{18} \text{ m}^3 \times (0.04/100) \\ &= 2.453 \times 10^{15} \text{ m}^3\end{aligned}$$



World energy consumption vs. CO₂

World energy consumption per year ~ 500 EJ
Around 80-85% of global energy is being obtained by burning fossil fuels

How much CO₂ is being generated annually?



Composition of crude oil by weight:

Element	Percentage by weight
Carbon	83-85%
Hydrogen	10-14%
Nitrogen	0.1 to 2%
Sulfur	0.05 – 6%

i.e., 400 kJ of energy is released through the formation of 1 mol of CO₂

How many moles of CO₂ would be formed to generate an energy of 500 EJ (5 * 10²⁰ J)?

No. of moles of CO₂ produced globally per year ~ 5 * 10²⁰ / 4*10⁵ = 1.25 * 10¹⁵ moles of CO₂



Estimation of volume of CO₂ produced globally per year

No. of moles of CO₂ produced globally per year $\sim 5 * 10^{20} / 4 * 10^5 = 1.25 * 10^{15}$ moles of CO₂

1 mol of CO₂ occupies 22.4 Liter volume at STP conditions

T = 0 °C; P = 1 atm

22.4 Liter = $22.4 * 10^{-3} \text{ m}^3$

$1.25 * 10^{15}$ moles of CO₂ occupies a volume of $\sim 22.4 * 10^{-3} * 1.25 * 10^{15}$
 $= 2.83 * 10^{13} \text{ m}^3$

How much time left for doubling the amount of CO₂ in the atmosphere?

Time required to double CO₂ in the atmosphere at the present rate of usage is :

- (a) Repeat the calculation at SATP
- (b) If the World annual energy consumption becomes three times that of the current consumption, Estimate the volume occupied by CO₂ in the atmosphere
- (c) What would be the safe estimate the volume of CO₂ in the atmosphere that can be manageable and accordingly estimate the share of fossil fuels towards energy generation worldwide annually?

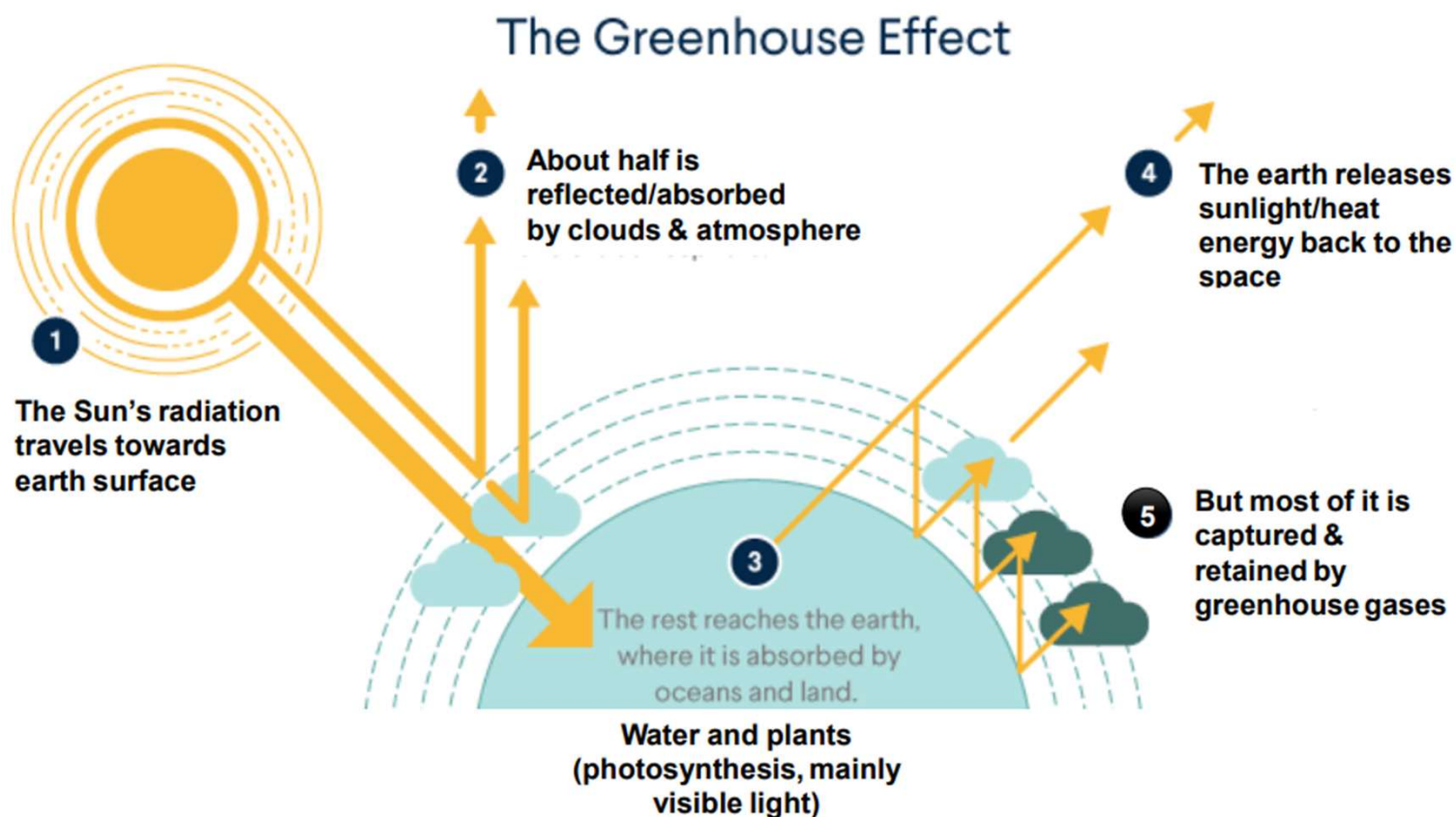
Does the greenhouse effect good or bad for earth and ecosystem?

- Earth's average temperature is 14 °C
- Mars's average temperature is -63 °C

“The greenhouse effect is a good thing. It warms the planet to its comfortable average of 14 °C and keeps life on earth and livable. Without it the world would be a frozen place, more like Mars. The problem is, mankind's voracious burning of fossil fuels for energy is artificially amping up the natural greenhouse effect. The result? An increase in global warming that is altering the planet's climate systems in countless ways.”

“Identified by scientists around 1850-1900 (**pre-industrial era**), the greenhouse effect is the natural warming of the earth that results when gases in the atmosphere trap heat from the earth that would otherwise escape into space.”

Greenhouse effect/global warming



Greenhouse/glasshouse



Glass house

- A house constructed by a transparent materials like glass and used for the cultivation or protection of tender plants, e.g., vegetables, flowers or fruits.
- The solar radiation passes through the glass is absorbed by the floor and contents, then the house become warmer. As the structure is not open to the atmosphere, the hot air cannot escape via convection, so the temperature inside the greenhouse rises.

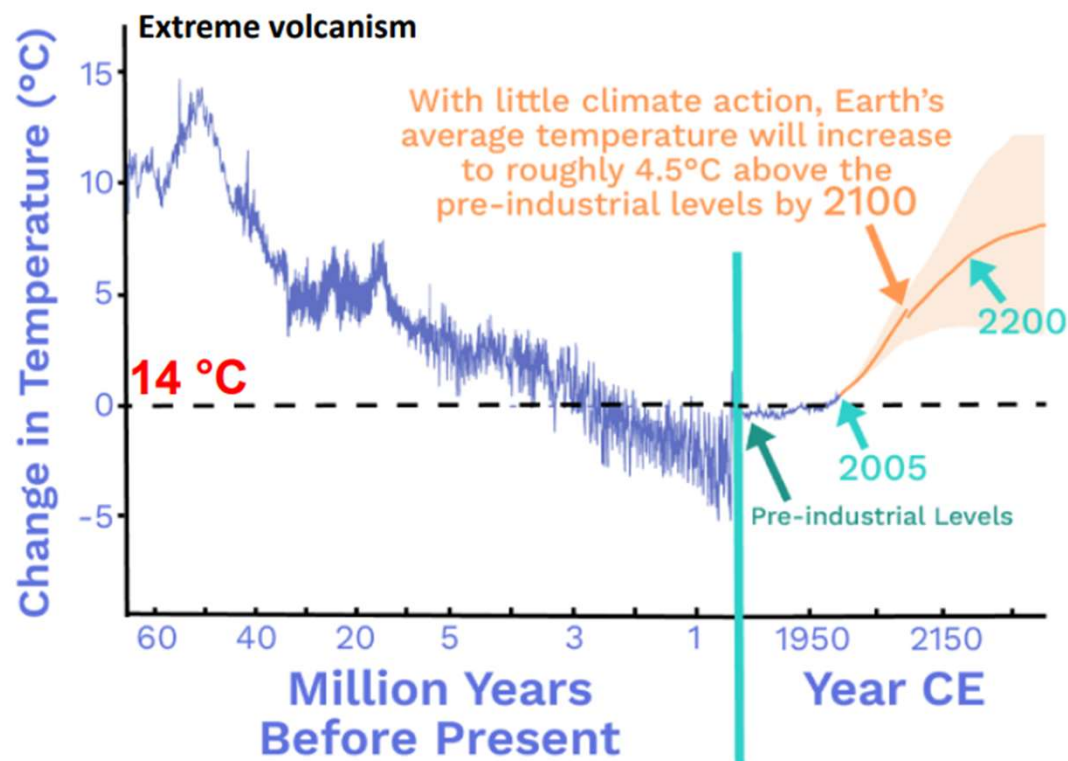


Greenhouse because

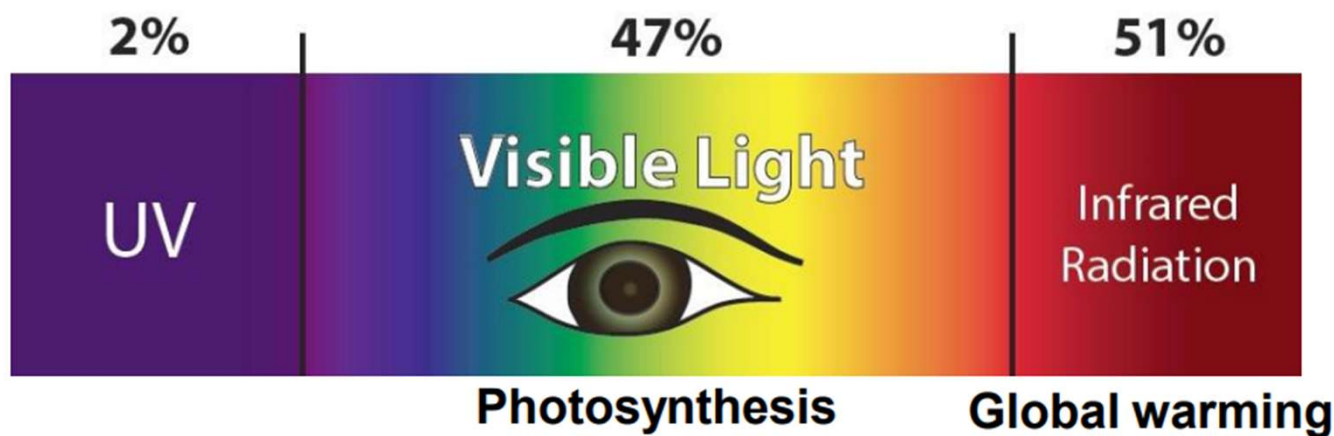
- It protects plants against excessive cold conditions.
- It prevents the accessibility of insects and animals that have the potential to damage or destroy plants.
- It reduces the use of pesticides.

Greenhouse effect/global warming

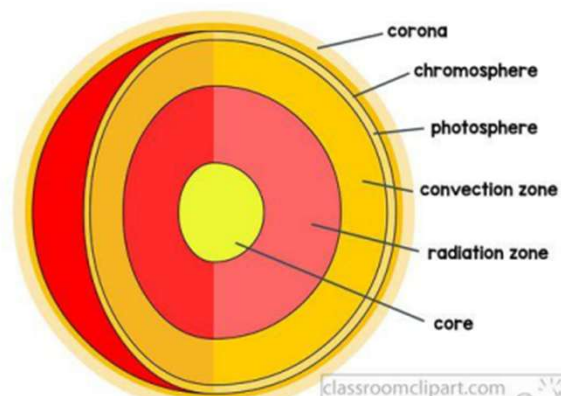
The global mean temperature in 2022 is currently estimated to be about 1.15 °C above the **1850-1900 pre-industrial average**.



Sunlight



- Infrared radiation from chromosphere
- UV and visible light from photosphere
- X-rays from corona mainly during solar flares



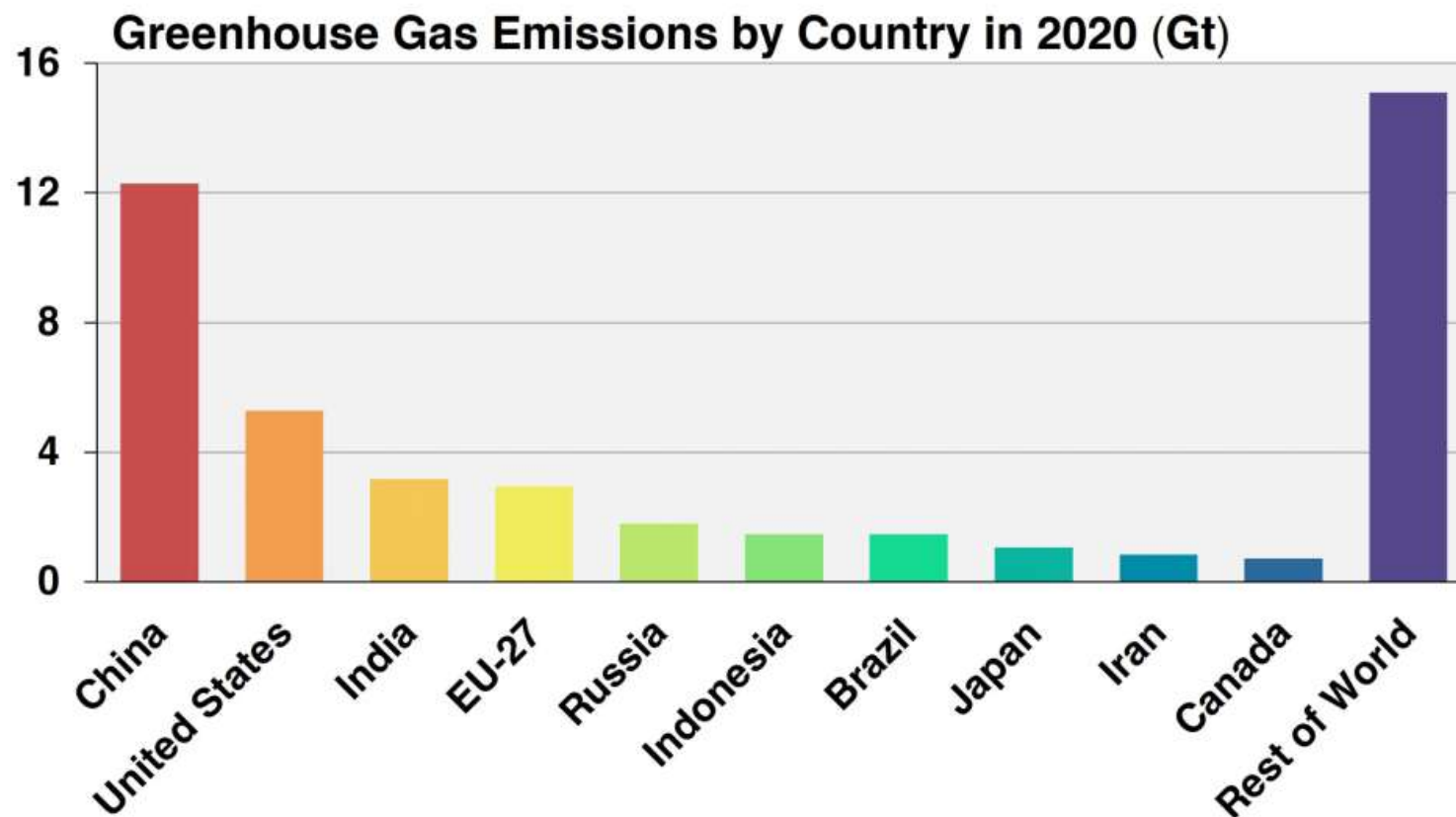
Scattering and absorption

- When sunlight hits the Earth's surface, some of it is reflected back towards space. **Greenhouse gases**, which allow the shorter wave radiation (visible light radiation) to pass through, which is **scattered by O_2 and N_2 molecules**, absorb longer wave radiation (infrared).
- **So visible light is scattered by O_2/N_2 and infrared light is absorbed by GHGs.** Light scattering/absorption is the property of the atom/molecule that purely depends on the molecular/atomic size with respect to wavelength of infrared/visible light.
- The **absorption of the infrared radiation** causes the molecules of the GHGs gases to vibrate from ground energy state to higher energy states and then **emission in the form of heat energy.**
- **Water vapor, carbon dioxide, methane, and other trace gases** absorb the longer wavelengths of outgoing infrared radiation.

Greenhouse emissions

- Water vapor
- Carbon dioxide (CO_2)
- Methane (CH_4)
- Nitrous oxide (N_2O)
- Exhaust particles (carbon and sulphate)
- Fluorine compounds
- Ozone (O_3) layer

Net GHG emissions



Source: Data from Climate Watch; Climate Watch Historical GHG Emissions (1990-2020). 2023.
Washington, DC: World Resources Institute. Available online at: https://www.climatewatchdata.org/ghg-emissions?end_year=2021&start_year=1990

Residence time and global warming potential

Residence time or life time: It is the average time it takes for a GHG molecule to be removed naturally.

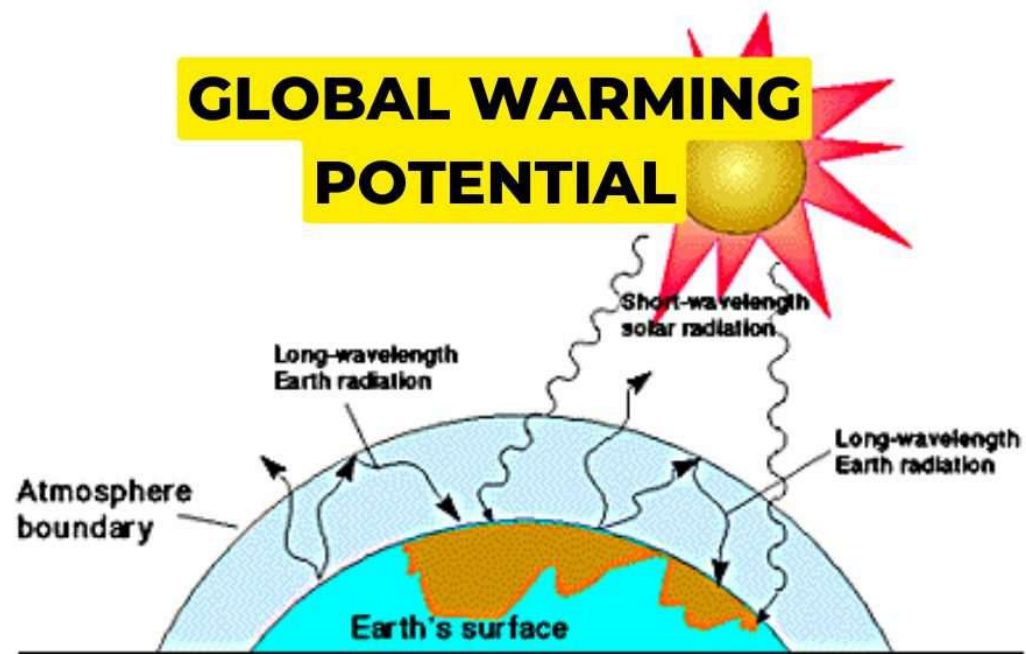
Methane: Every molecule of methane that goes into the atmosphere remains there for **12 years** until it is removed by oxidation (sunlight) into carbon dioxide (CO_2) and water (H_2O).

Nitrous oxide: It is removed from the atmosphere (**120 years**) when it is destroyed by ultraviolet radiation or photochemical reactions.

Global warming potential (GWP): It is used to compare the contributions of the various greenhouse gases. It is the heat absorbed by any greenhouse gas in the atmosphere over a period of time, as a multiple of the heat that would be absorbed by the same mass of carbon dioxide. **GWP is 1 for CO_2 .**

Gas	GWP	Lifetime (Y)
Carbon dioxide (CO_2)	1	Variable
Methane (CH_4)	25	12.2
Nitrous oxide (N_2O)	298	120
Hydrofluorocarbons (HFCs)	12-14,800	1.5-264
Perfluorocarbons (PFCs)	7,390-12,200	3200-50000
Sulphur hexafluoride (SF_6)	22,800	3200

Human activity caused global warming
Magnitude of global warming in the future depends on how seriously we choose to limit GHGs



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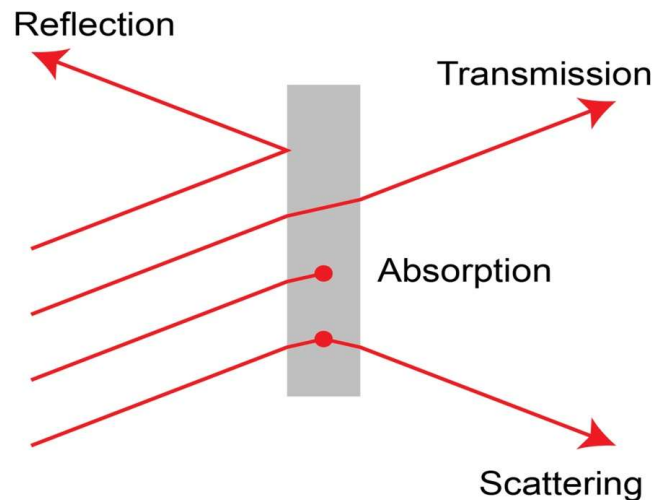
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Scattering and absorption

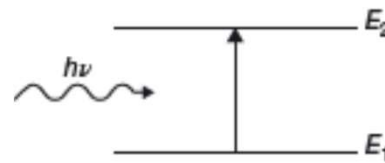
- Visible light is mostly **scattered** by O₂ and N₂ and infrared light is **absorbed** by GHGs.
- When the light change its direction without a change in energy/frequency, then it is called **Rayleigh scattering**, but sometimes the light can disappear with the energy transferred to the object, then it is called **absorption**.



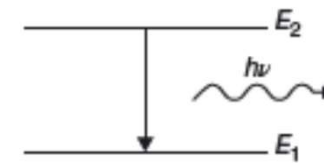
$\Delta E = E$ difference between energy levels

$$\Delta E = E_2 - E_1$$

$$E = h\nu = \frac{hc}{\lambda}$$



Absorption



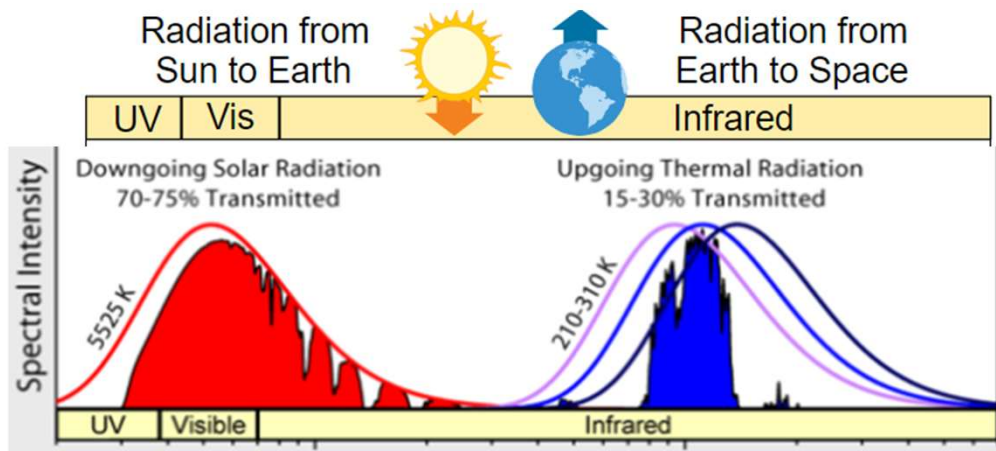
Emission

Wavelength of Vis $\ll$ IR, **because of smaller wavelength (smaller particles) it can change its direction**



Atmospheric window

An **atmospheric window** is the portion of the electromagnetic spectrum that can be transmitted through the atmosphere.



Earth's Energy Balance:

The infrared atmospheric window plays a crucial role in Earth's energy balance by allowing thermal radiation from the surface to escape into space, which helps regulate the planet's temperature.

- The filled-in red area is the spectrum of the sunlight that reaches the earth and the missing radiation (25-30%) is either absorbed by GHGs or scattered by O_2/N_2 molecules.
- The filled-in blue area is the spectrum of the sunlight that released/reflected from earth to space and the remaining radiation (70-85%) is absorbed by GHGs.



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Carbon dioxide (CO_2) is by far the most important greenhouse gas. Analysis of **ice cores in Greenland and Antarctica** shows that **pre-industrial (1850-1900)** levels of carbon dioxide in the atmosphere were about **280 ppmv** (parts per million by volume). The CO_2 atmospheric level in 2022 was **420 ppm**.

Fossil fuel combustion (coal, crude oil, and methane), **cement production** and other industrial processes are the major source of CO_2 emissions.

Cement industry: CaO is a key ingredient in cement. CaO is produced from CaCO_3 decomposition with the release of CO_2 .



Changes in land use (mainly **deforestation**) also contribute to carbon dioxide emissions. They represent slightly less than a fifth of the total emissions of carbon dioxide.



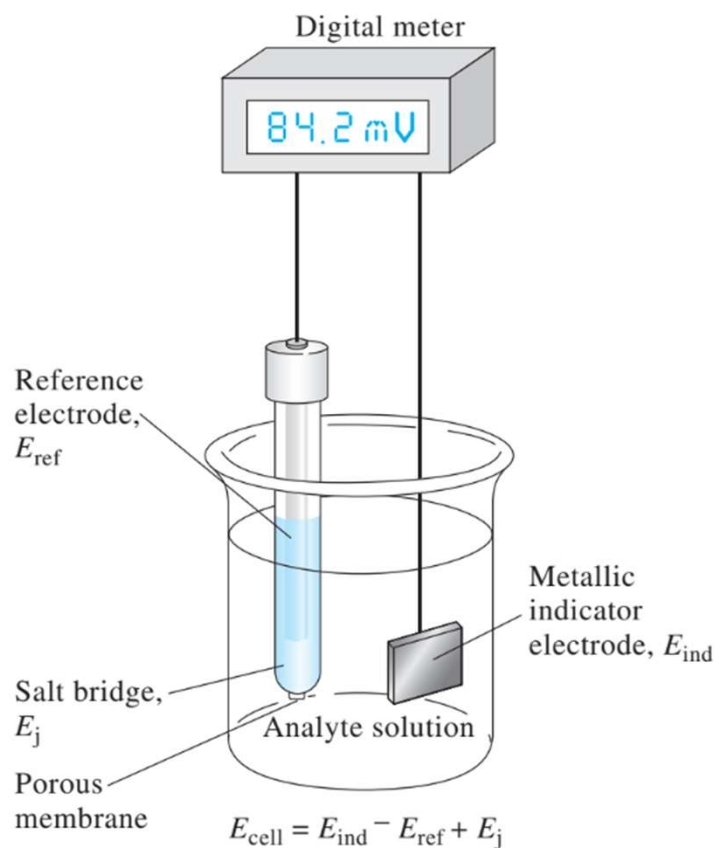
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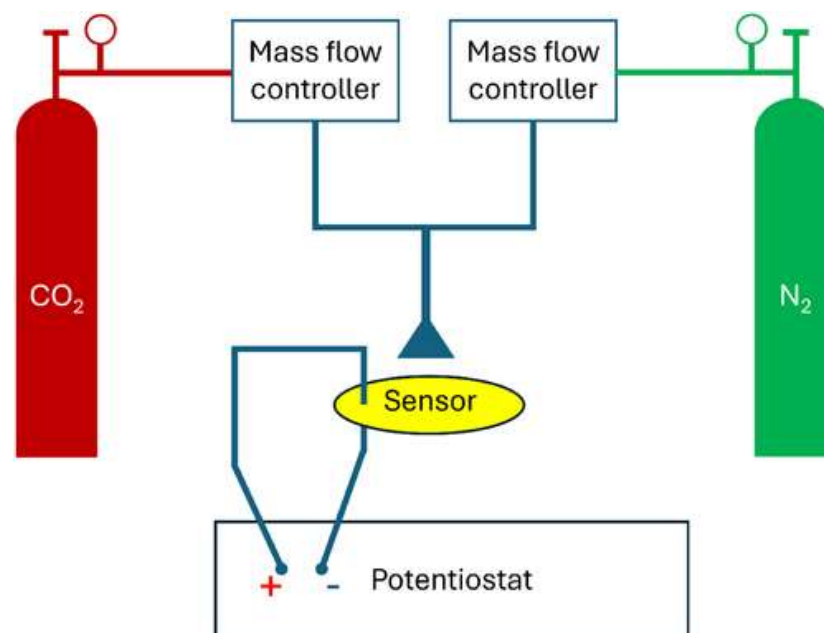
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Potentiometric Determination of CO₂



A gas-sensing probe is a galvanic cell whose potential is related to the concentration of a gas in a solution.



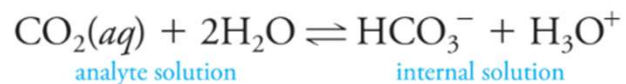
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Potentiometric Determination of CO₂

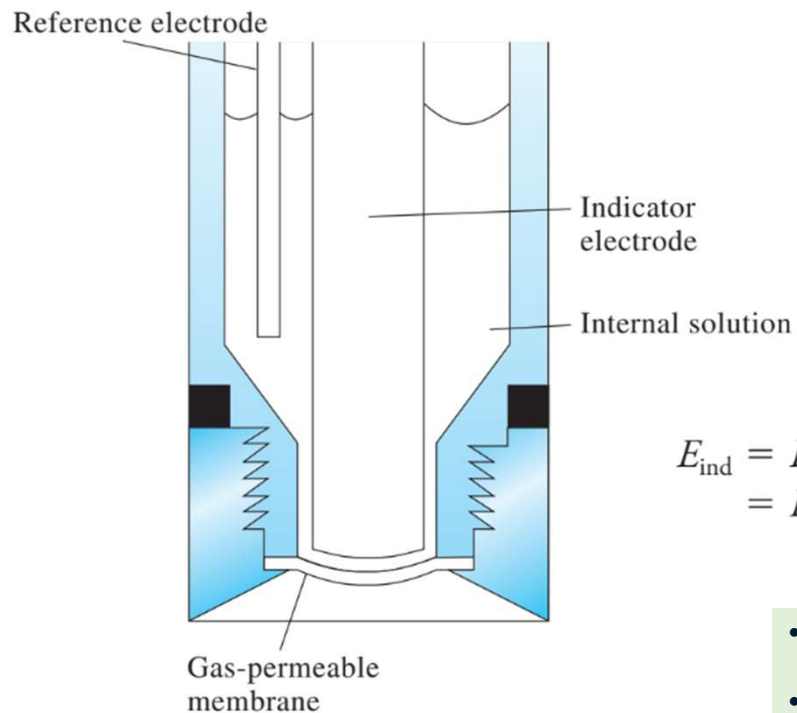


$$K = \frac{(a_{\text{H}_3\text{O}^+})_{\text{int}}(a_{\text{HCO}_3^-})_{\text{int}}}{[\text{CO}_2(aq)]_{\text{ext}}}$$

$$(a_{\text{H}_3\text{O}^+})_{\text{int}} = a_1 = K_g[\text{CO}_2(aq)]_{\text{ext}}$$

$$\begin{aligned} E_{\text{ind}} &= L + 0.0592 \log a_1 = L + 0.0592 \log K_g[\text{CO}_2(aq)]_{\text{ext}} \\ &= L + 0.0592 \log K_g + 0.0592 \log [\text{CO}_2(aq)]_{\text{ext}} \end{aligned}$$

- Monitoring CO₂ levels in industrial processes.
- Measuring CO₂ in food packaging to assess freshness.
- Determining total organic carbon content in soil.



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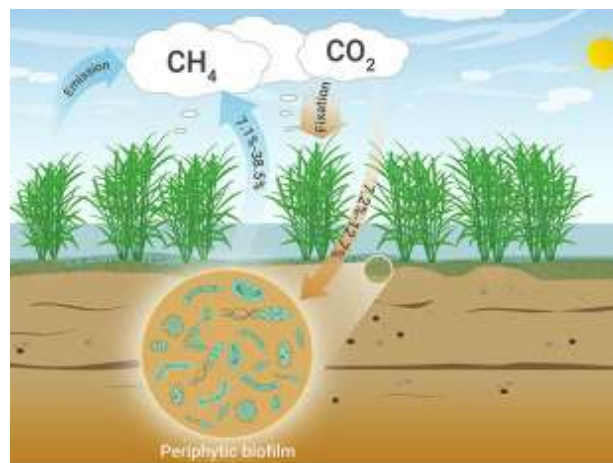
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Methane (CH₄)

- It is formed naturally by the bacterial decomposition of organic matter under oxygen-free conditions in soil.
- **Rice cultivation** and **emissions from coal mines** represent significant sources of methane emissions.
- The pre-industrial concentration of methane is estimated to be **0.7 ppm**. Today's level is more than twice as high, about **1.8 ppm**.



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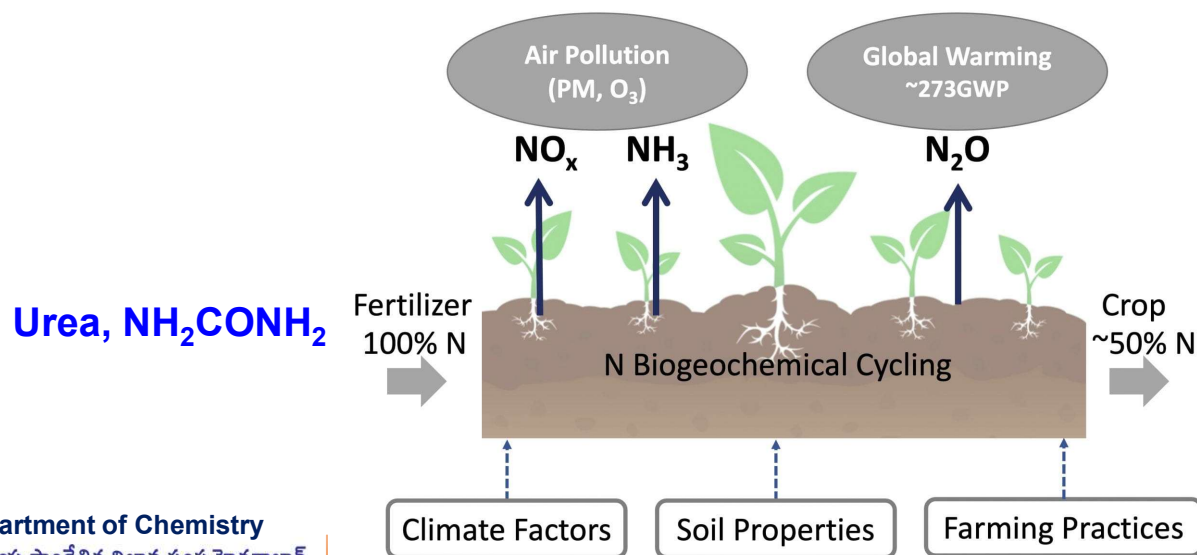
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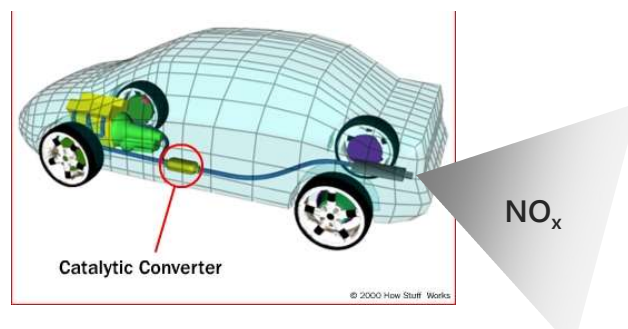
Nitrous oxide (N_2O , laughing gas)

- Denitrification is the main source of nitrous oxide in the atmosphere.
- Denitrification is the microbial process of reducing nitrate and nitrite in the soil to gaseous forms of nitrous oxide (N_2O) and nitrogen (N_2). This process, which is carried out by micro-organisms, occurs naturally in the soil.
- The more nitrogen is made available to plants by adding it in the form of fertilizers, the more nitrous oxide is formed.



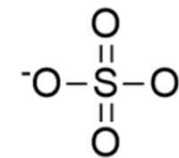
Nitrous oxide (N_2O , laughing gas)

- Another source of nitrous oxide emissions is all sorts of **combustion**. During the combustion process, **small amounts of N_2O are formed in addition to the “ordinary” nitrogen oxides (NO and NO_2) via reaction of O_2 with N_2 present in air.** This amount depends largely on the combustion conditions.
- Nitrous oxide is a greenhouse gas whose pre-industrial (**1850-1900**) level is estimated to have been **270 ppb (parts per billion)**. The level in **2021 was 340 parts per billion**, an increase of 20 percent.
- Nitrous oxide has a long residence time in the atmosphere, an average of about 120 years.”



Particles – global warming

- Particles like **sulfate and carbon** (known as particulate matter) in the atmosphere also affect the radiation balance and global warming.

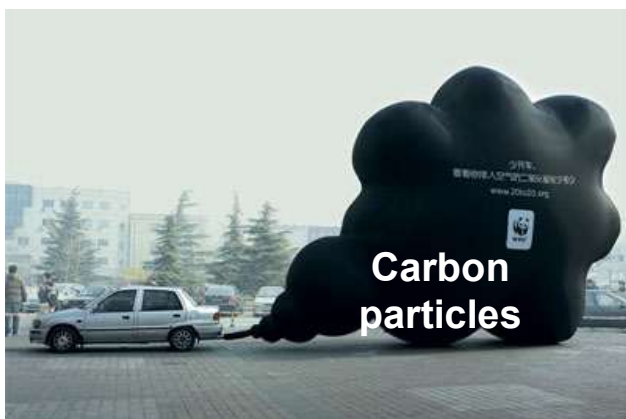


- Sulphate particles can absorb IR light strongly.
- **Sulphate particles** are formed by the oxidation of sulfur dioxide (SO₂) gas in the atmosphere.
- SO₂ is emitted largely in fossil fuel (crude oil and coal) combustion, particularly from **power plants burning coal** (0.5% to 5% sulphur).
- **Acid rain** results when SO₂ with water and O₂ in the atmosphere to form sulfuric acid (H₂SO₄).



Carbon particles from diesel engines

- There are also carbon particles in the air, mainly released from the diesel engines.
- These carbon particles can both absorb heat and reflect incoming light.
- Unlike other greenhouse gases, the residence time of particles in the air is short, **about two weeks**.



Diesel engine vehicles



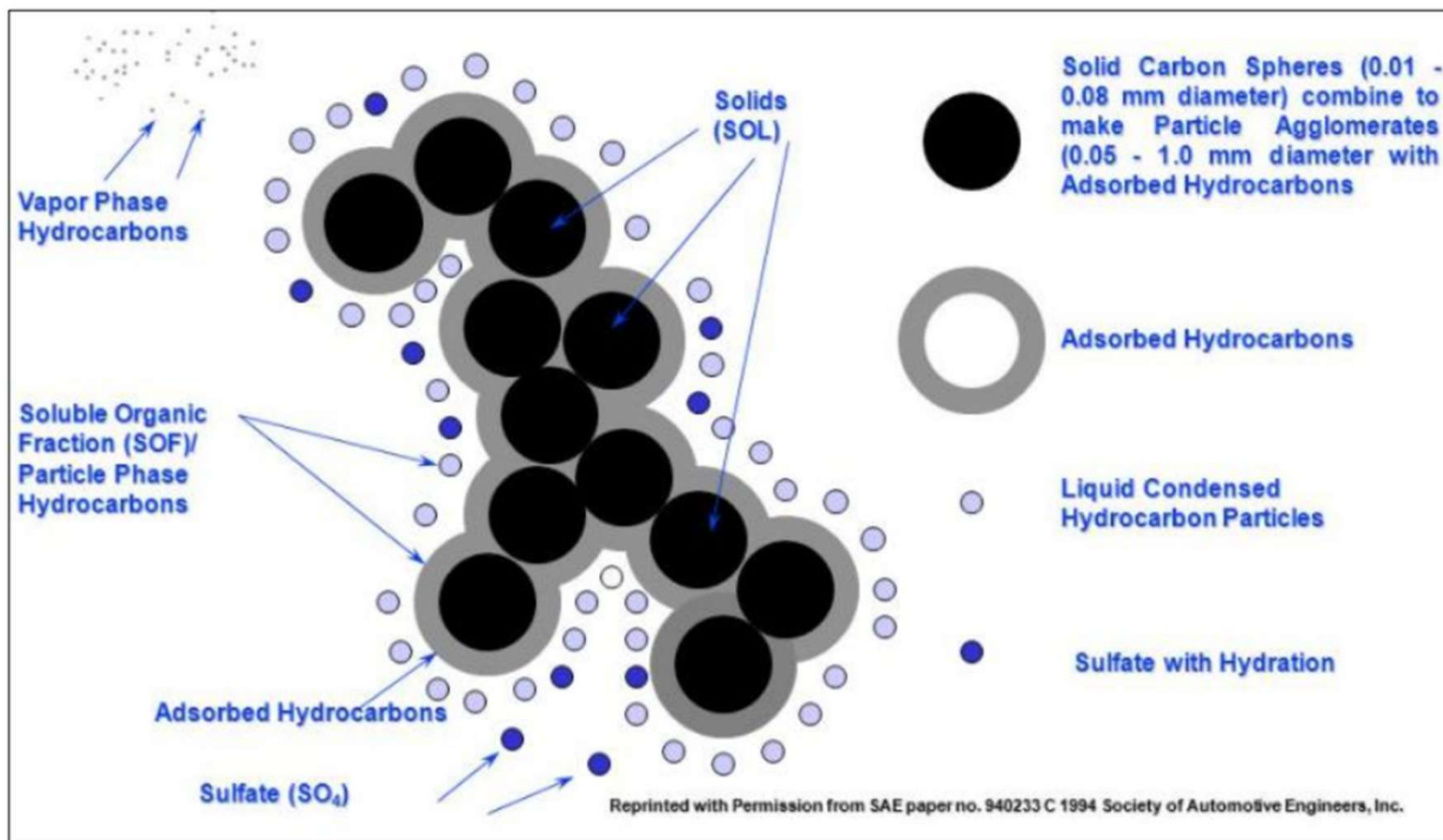
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Carbon particles from diesel engines



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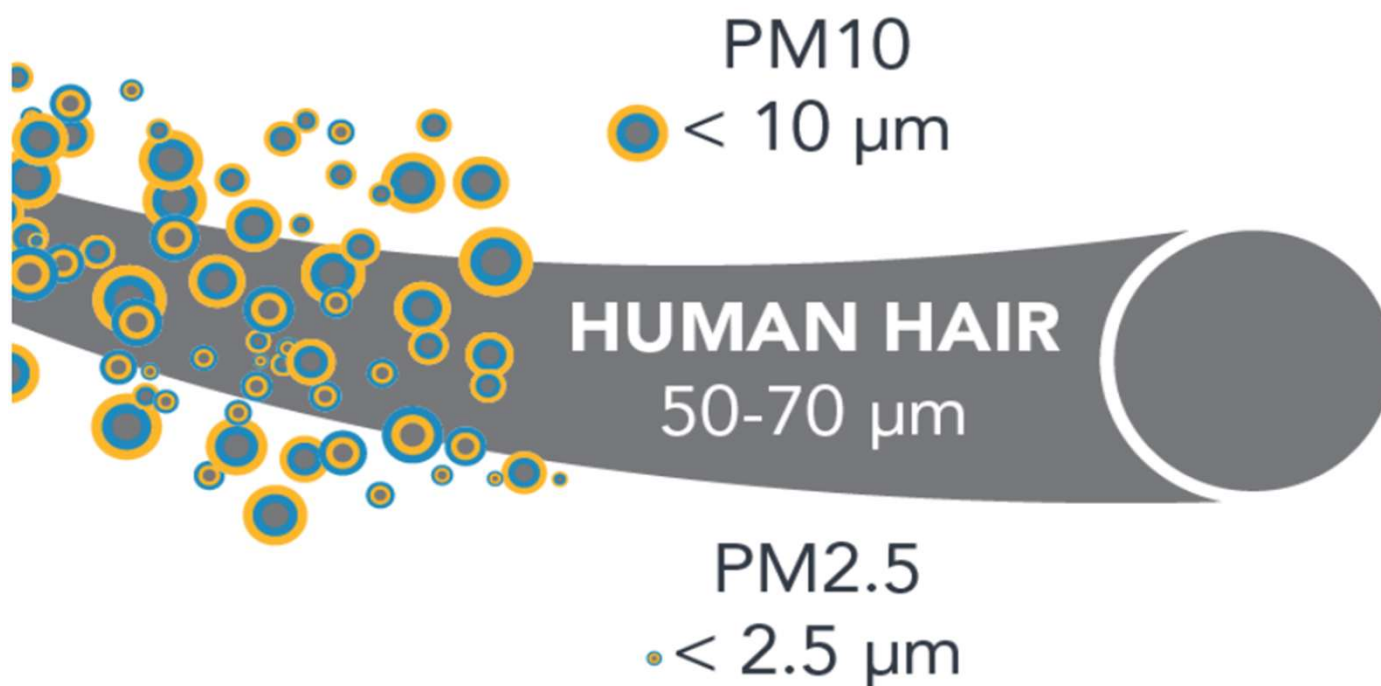
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Particulate matter, PM: Asthma disease

Particulate Size Comparison



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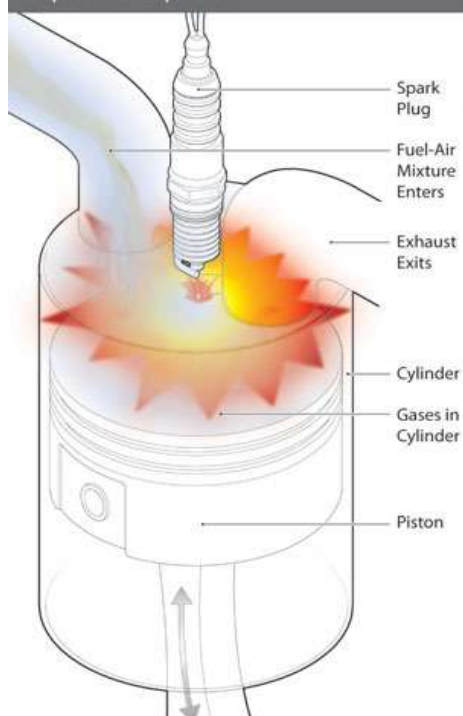
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Carbon particles from diesel engines

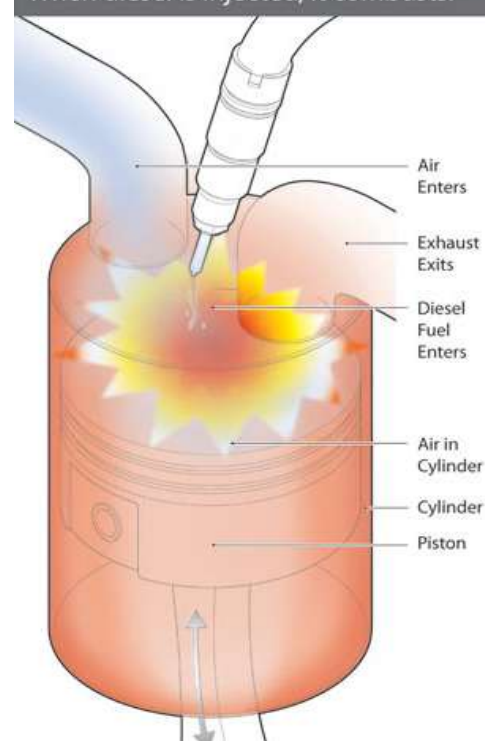
Petrol engine

GAS ENGINE: Gasoline does not self-ignite when compressed. It needs a spark to explode.



Diesel engine

DIESEL ENGINE: A diesel engine compresses (heats) air in the cylinder. When diesel is injected, it combusts.



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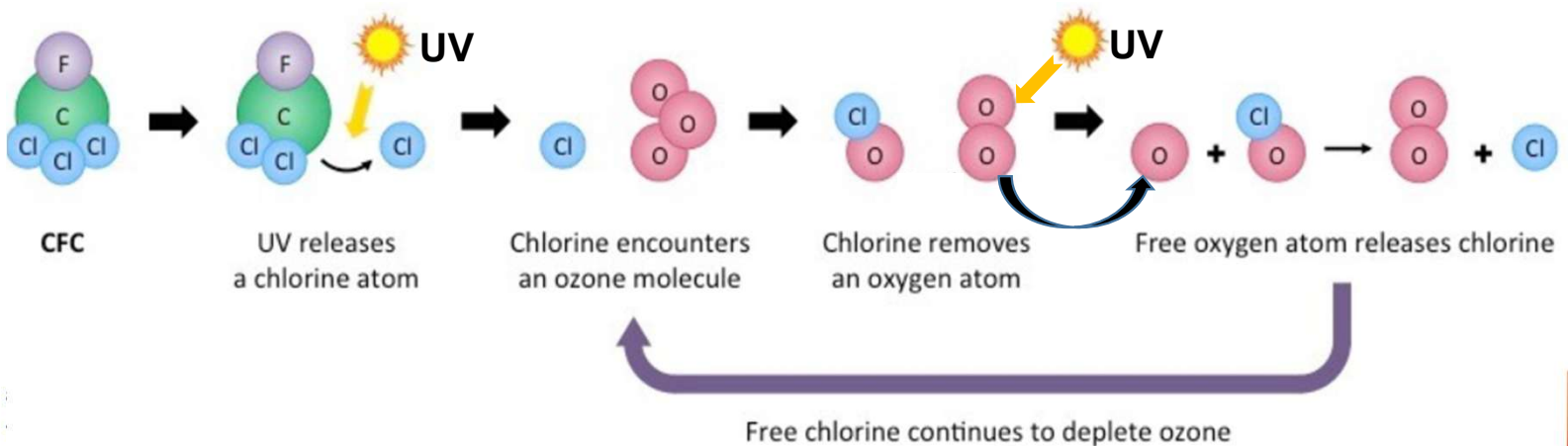
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Fluorinated compounds (CFCs)

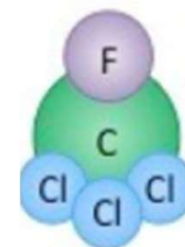
- **Chlorofluorocarbons (known as CFCs)**, which have mainly attracted attention because of their ability to break down the stratospheric ozone.
- The important property of CFCs is their high volatility, having boiling points close to zero degrees centigrade.
- **Refrigerators, air conditioners, sprays, and fire extinguishers.**
- CFCs circulates in the walls of the fridge, absorbing the ambient heat to keep the fridge cooled down. As CFCs absorb the heat, it evaporates into the atmosphere.



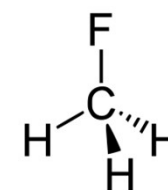
Fluorinated compounds (HCFC and HFC)

- CFCs absorb the infrared radiation (and trap the heat) very efficiently, because of the nature of **the carbon-halogen bonds**.
- Because of their effect on the ozone depletion, CFC gases are being replaced by HCFC and HFC gases.

CFC	→	Chlorofluorocarbon	
		↓ ↓ ↓	
		Chlorine Fluorine Carbon	
HCFC	→	Hydrochlorofluorocarbon	
		↓ ↓ ↓ ↓	
		Hydrogen Chlorine Fluorine Carbon	
HFC	→	Hydrofluorocarbon	
		↓ ↓ ↓	
		Hydrogen Fluorine Carbon	



CFC



HFC



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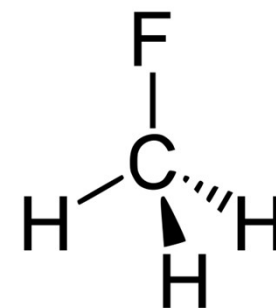
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Hydrofluorocarbons (HFCs)

- Hydrofluorocarbons (HFCs), which are similar to CFCs do not contain chlorine and therefore do not affect the ozone layer, but causing global warming because they absorb IR radiation.
- The breakdown of HFCs occurs in the troposphere, where they are split by reactions with hydroxyl radicals ($\cdot\text{OH}$).
- Because they contain hydrogen, HCFCs break down more easily in the atmosphere than do CFCs.
- CFCs do not easily react with other substances. In fact, they break up only by UV light, causing the release of chlorine (Cl).



HFC



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Perfluorocarbons (PFC)

- Perfluorocarbons (PFCs) are byproducts during [aluminum \(Al\) production](#) and semiconductor manufacturing
- Examples: tetrafluoromethane (CF_4) and hexafluoroethane (C_2F_6).

CFC	→	Chlorofluorocarbon ↓ ↓ ↓ Chlorine Fluorine Carbon
HCFC	→	Hydrochlorofluorocarbon ↓ ↓ ↓ ↓ Hydrogen Chlorine Fluorine Carbon
HFC	→	Hydrofluorocarbon ↓ ↓ ↓ Hydrogen Fluorine Carbon
PFC	→	Perfluorocarbon ↓ ↓ ↓ (only) Fluorine Carbon



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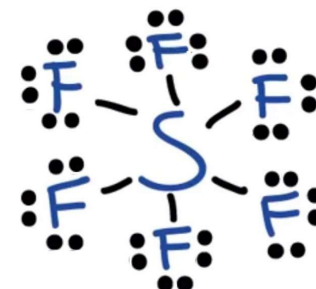
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Fluorinated compounds (SF₆)

- Sulphur hexafluoride (SF₆) is the most harmful greenhouse gas with a GWP 22,800 times the baseline of CO₂.
- In SF₆, all of the electrons of sulfur are paired giving a great stability to the molecule, thus high residence time.
- It is used mainly as an insulating gas in the electronics industry to control the high voltages in electrical equipment.
- Because of its wide applicability, large amounts of SF₆ have been leaked into the atmosphere.

Sulfur Hexafluoride



Gas	GWP	Lifetime (Y)
Carbon dioxide (CO ₂)	1	Variable
Methane (CH ₄)	25	12.2
Nitrous oxide (N ₂ O)	298	120
Hydrofluorocarbons (HFCs)	12-14,800	1.5-264
Perfluorocarbons (PFCs)	7,390-12,200	3200-50000
Sulphur hexafluoride (SF₆)	22,800	3200



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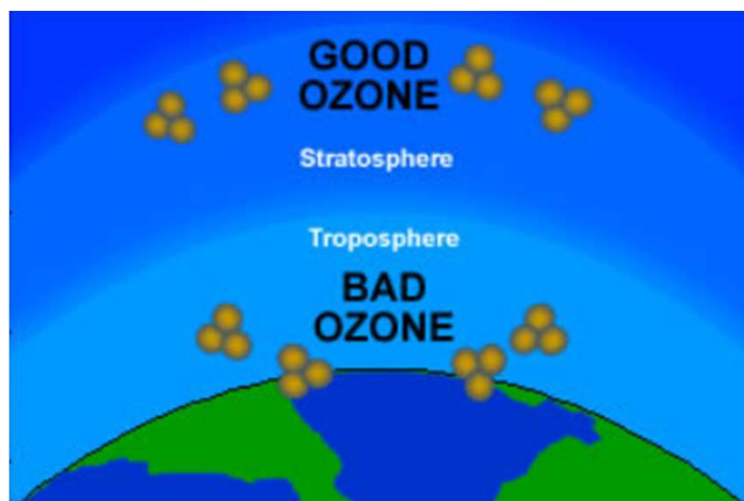
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Ozone (O_3) as a greenhouse gas

Atmospheric ozone has two effects

- It absorbs solar ultraviolet radiation in the stratosphere (**stratospheric ozone**).
- It also absorbs infrared radiation emitted by the Earth's surface, effectively trapping heat in the troposphere (**tropospheric ozone**).



Stratospheric or good ozone is formed in the stratosphere naturally due to **UV radiation breaking up O_2 molecules to oxygen radicals**, which reacts with available O_2 , forming ozone.

Tropospheric or bad ozone, is not emitted directly into the air, but is formed by chemical reactions between nitrogen oxides (**NO_x**) and volatile organic compounds (**VOC**) in the presence of sunlight.



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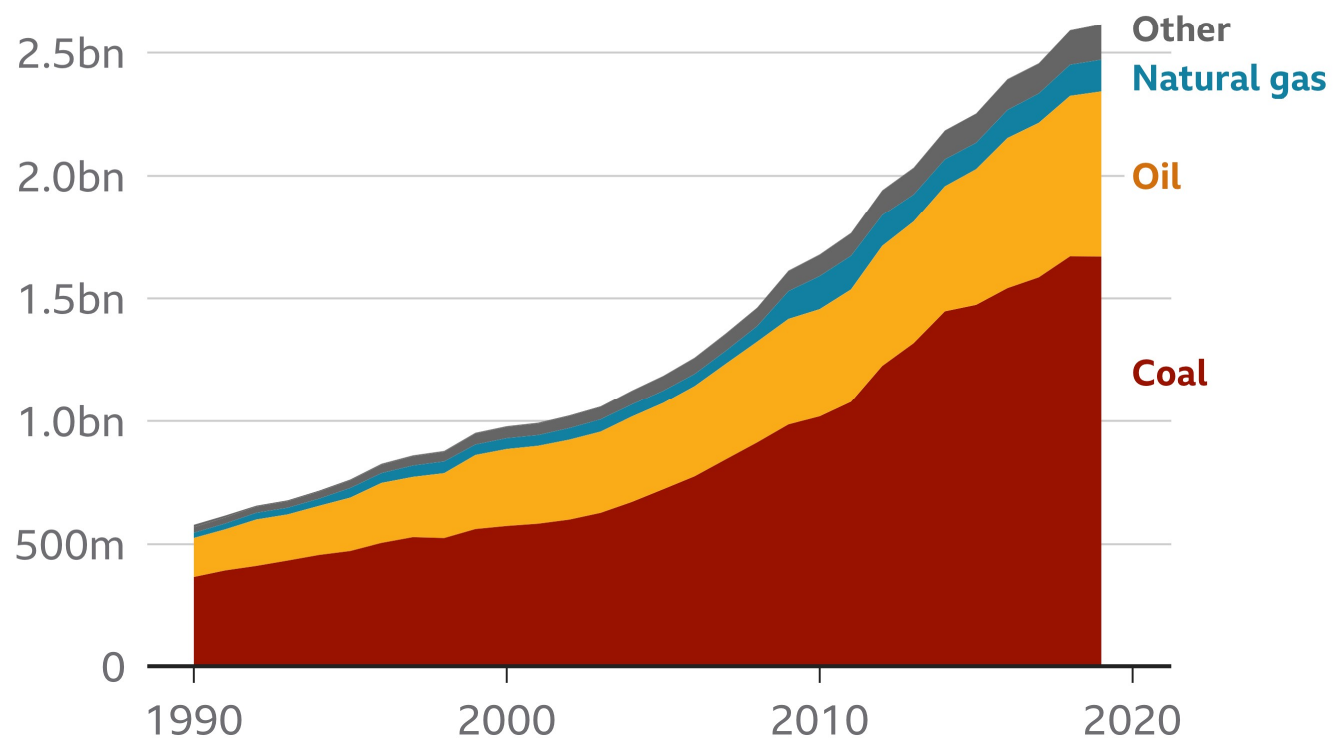
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India emissions

CO2 emissions in India by fuel type (billion tonnes)



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Climate Change mitigation

PARIS CLIMATE AGREEMENT



2015



Goal: The Paris Agreement's long-term temperature goal is to keep the rise in mean global temperature to well below 2°C (3.6°F) above pre-industrial levels, and preferably limit the increase to 1.5°C (2.7°F)



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[Paris Agreement](#)". United Nations Treaty Collection. [Archived](#) from the original on 5 July 2021. Retrieved 15 July 2021.

Solutions to control GHG emissions

Prevention

Cut fossil fuel use
(especially coal)

Shift from coal to natural gas

Improve energy efficiency

Shift to renewable energy
resources

Transfer energy efficiency
and renewable energy
technologies to
developing countries
Reduce deforestation

Use more sustainable
agriculture and forestry

Limit urban sprawl

Reduce poverty

Slow population growth



Cleanup

Remove CO₂ from air,
smokestack and vehicle
emissions

Store (sequester) CO₂ by
planting trees

Sequester CO₂ deep
underground (with no leaks
allowed)

Sequester CO₂ in soil by
using no-till cultivation and
taking cropland out of
production

Sequester CO₂ in the deep
ocean (with no leaks
allowed)

Repair leaky natural gas
pipelines and facilities

Use animal feeds that
reduce CH₄ emissions from
cows (belching)



Department of

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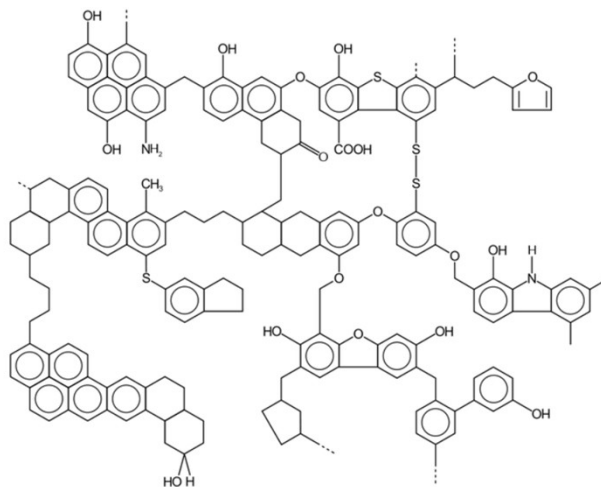
Shifting from coal to natural gas

- **Methane (natural gas) is the simplest hydrocarbon (CH_4) gas, so it burns more efficiently than coal**, making it a better option than coal.
- **Methane contains a single carbon atom.** As a result, methane emits lower amount of CO_2 than other fossil fuels.
- Electricity generation from efficient natural gas plants emits roughly **60% less CO_2** per unit of electrical energy than generation from coal power plants.
- Because of its simple structure, methane as a fuel emits much less **particulate matter (carbon particles)** than coal.
- **Coal contains sulphur (0.5% to 5%), so burning coal for power plants emits considerable amount of SO_2** which is responsible for acid rain and global warming (sulphate particles).



Shifting from coal to natural gas

- Coal has a complex polymeric structure, mainly contains carbon (70- 90%), along with hydrogen and oxygen species.
- **It also contains sulphur (0.5% to 5%),** so burning coal emits SO_2 which is responsible for acid rain and sulphate particles (global warming).
- **Coal also contains nitrogen (0.5-3%),** so considerable amount of NO_x pollutants are emitted via coal burning.



Typical chemical structure of coal



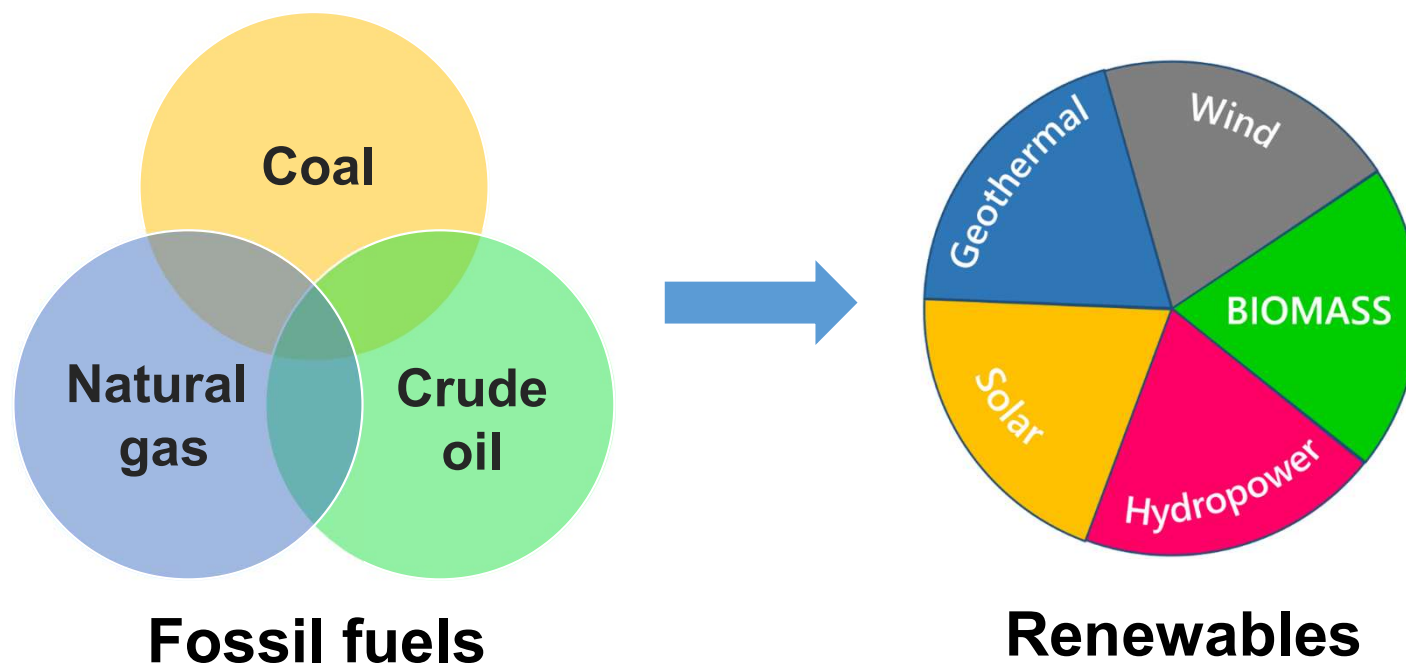
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Shifting to renewable energy



CO₂, SO₂, NO_x, carbon particles, etc.



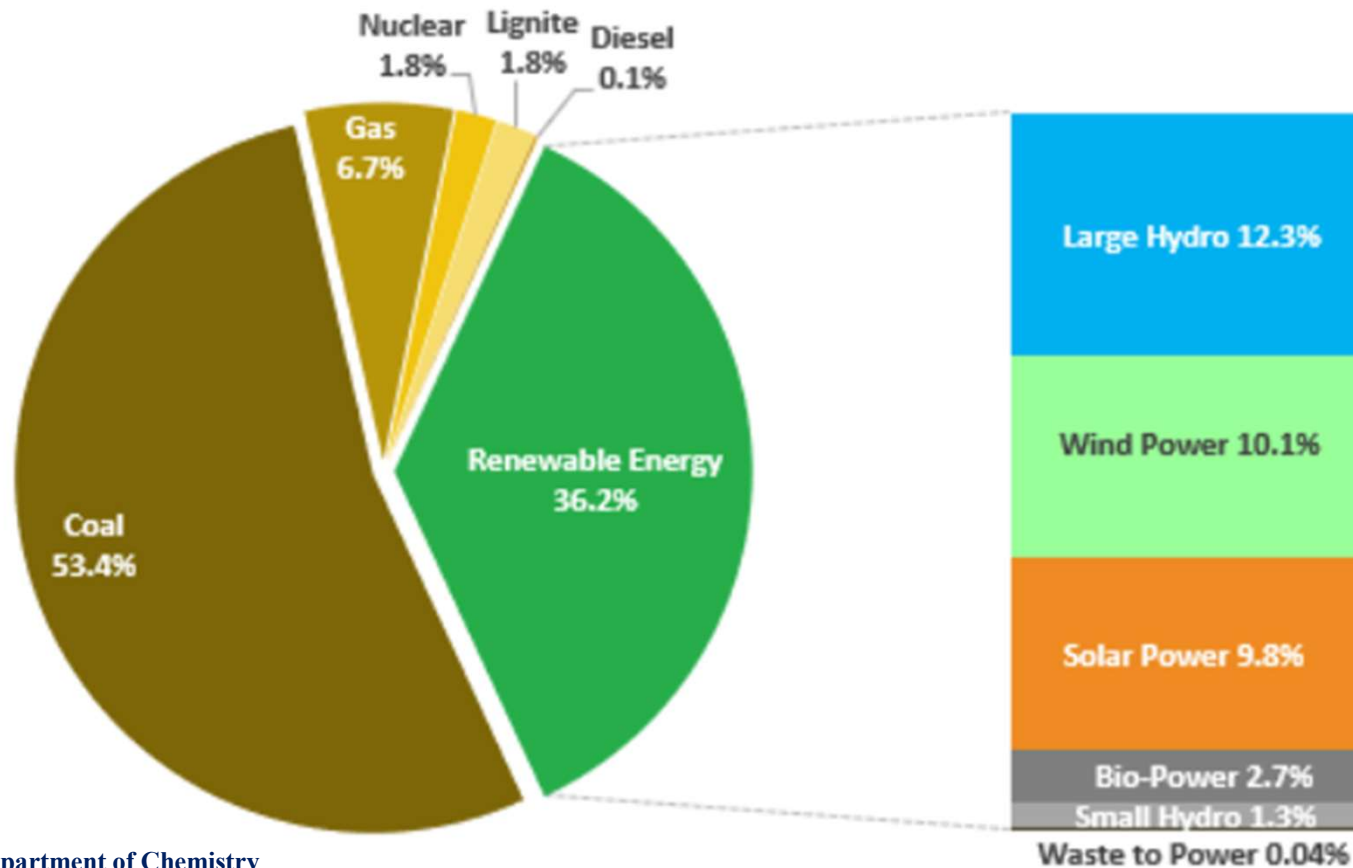
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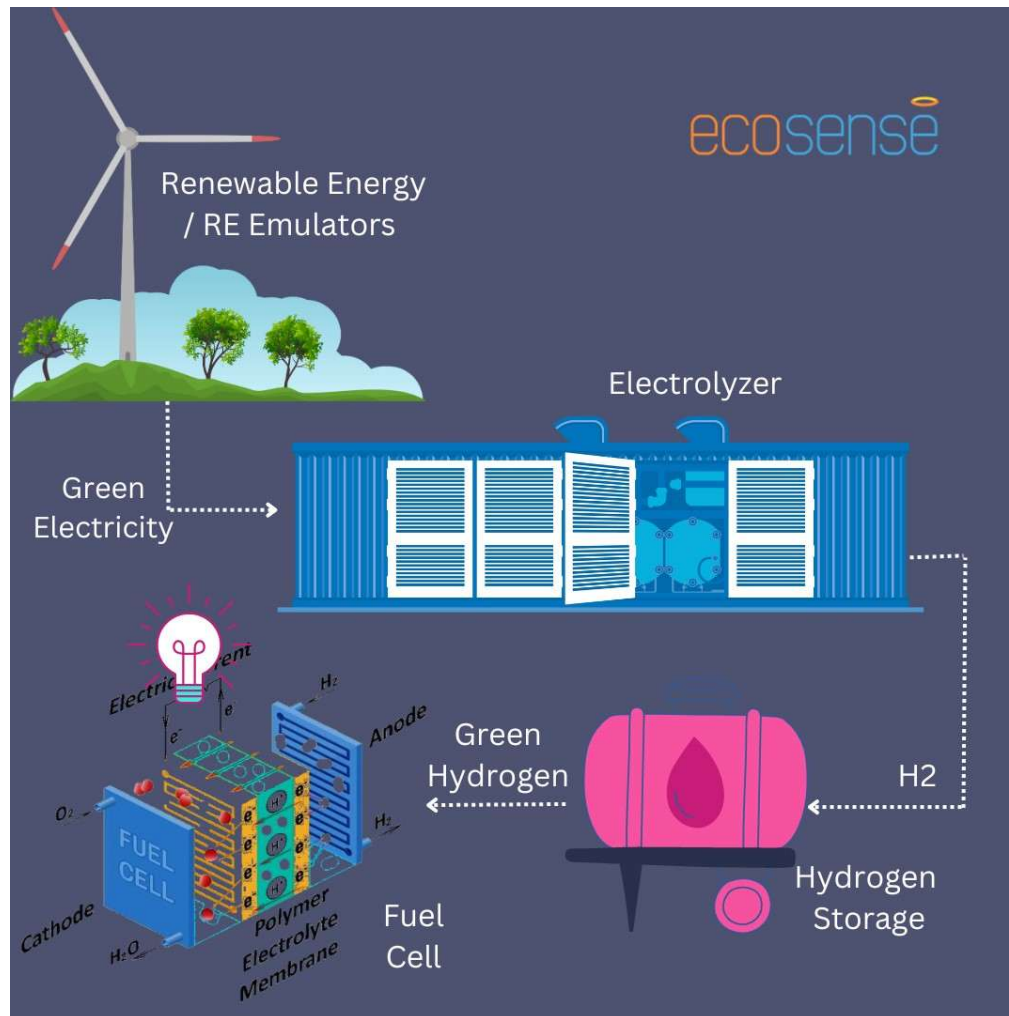
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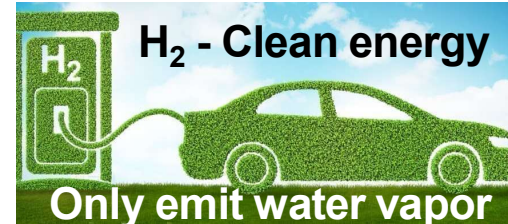
Where does India stand in the energy transition?



Green H₂ for electrical vehicles



Types of H₂



Grey H₂

Steam methane reforming (50% H₂)
CO₂ is emitted into the atmosphere



Black/brown H₂

Gasification of coal (20% H₂)
CO₂ is emitted into the atmosphere



Blue H₂

Steam methane reforming
CO₂ is captured & stored

Green H₂

Water electrolysis (4% H₂)
Green electricity from solar/wind



Plastic Pollution

Economic consideration
Number of times recyclable
Any pollutants emerge?

The 7 Types Of Plastics

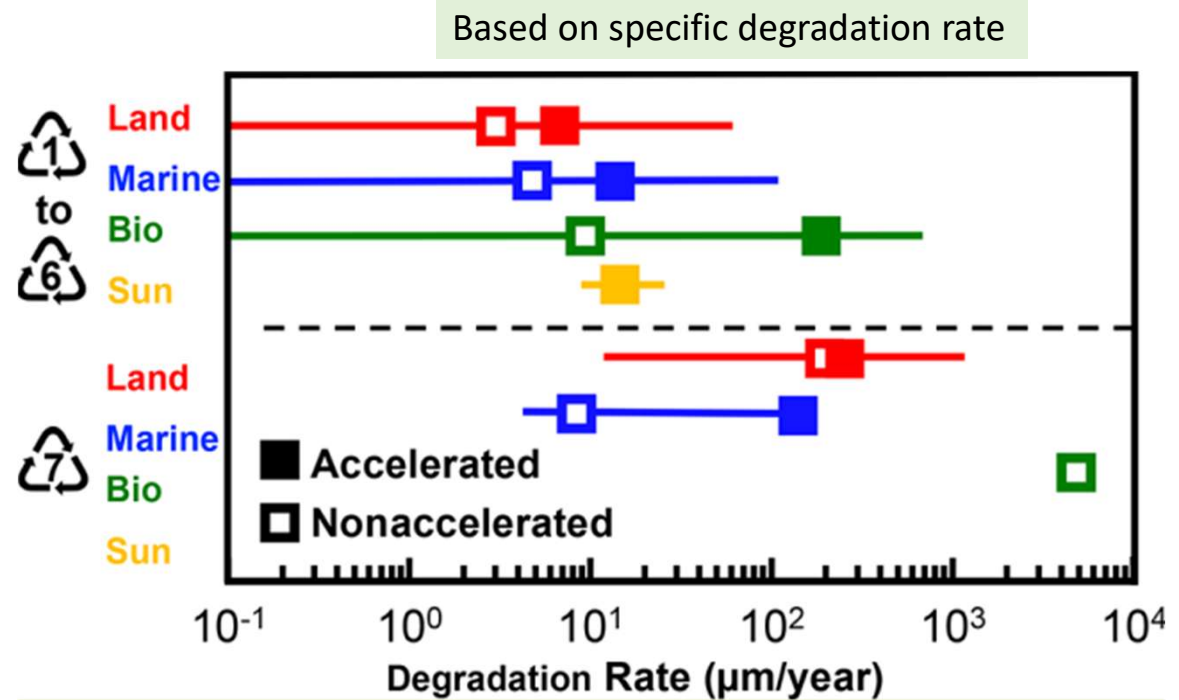
 PETE	 HDPE	 PVC	 LDPE	 PP	 PS	 Other Plastics
Polyethylene terephthalate	High-density polyethylene	Polyvinyl chloride	Low-density polyethylene	Polypropylene	Polystyrene or Styrofoam	Other plastic including polycarbonate & biodegradable plastic
soda bottles, water bottles, peanut butter jars, salad dressing bottles, medicine containers and vinegar bottles	milk jugs, laundry detergent bottles, shampoo/conditioner bottles, and bleach bottles	pipes, shower curtains, clear medical tubing, vinyl records, cooking oil bottles, seat covers, and coffee containers	sandwich bags, shrink wrap, grocery bags, squeezable condiment bottles and bread bags	yogurt cups, ketchup bottles, syrup bottles, plastic bottle caps and 'microwave-safe' plastic containers	disposable cups, take-out food containers, packing peanuts, egg cartons and Styrofoam insulation	baby bottles, sippy cups, water cooler bottles, polycarbonate plastic food containers, and car parts
						



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Fate of polymers under various environmental conditions

Plastics type	Common applications
1. PET	Single-use water bottle
2. HDPE	Plastic bottles
2. HDPE	Pipes
3. PVC	Pipes
4. LDPE	Plastic bags
5. PP	Food storage container
6. PS	Insulating packaging
7. Others	Biodegradable plastic bag



The relative advantages and disadvantages of dematerialization, substitution, reuse, material recycling, waste-to-energy, and conversion technologies must be carefully considered to design the best solutions to the environmental challenges posed by the enormous and sustained global growth in plastics production and use.



Solution to plastic pollution

Plastic has slowly become an integral part of all human requirements. Plastic carry bags, packaging material, bottles, cups, and various other items have slowly replaced everything made of other material due to the advantages of plastic.

Plastic is durable, easy to produce, lightweight, unbreakable, odorless, and chemical resistant.

Disadvantages

- Leaching of microplastics into soil and water bodies
- Accumulation by aquatic life

Prof. R. Vasudevan



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http://timesofindia.indiatimes.com/articleshow/84883086.cms?utm_source=contentofinterest&utm_medium=text&utm_campaign=cppst

Plastic-to-fuel Technology

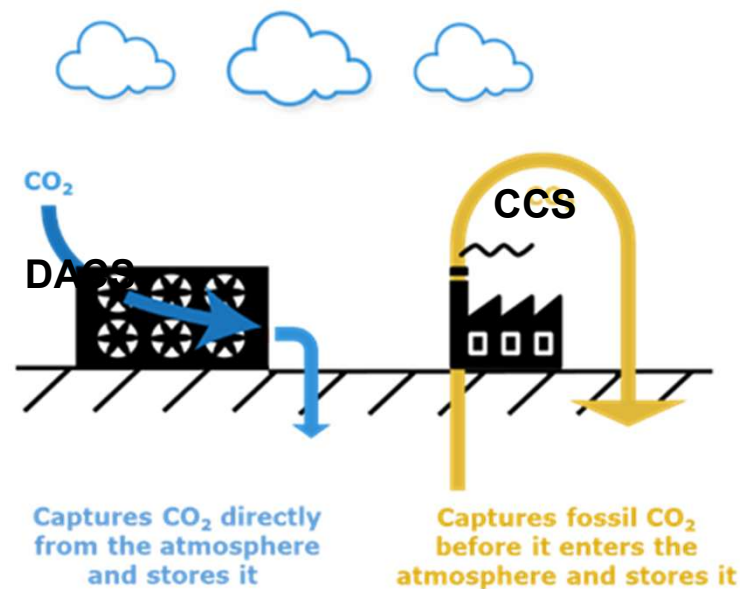
- Pyrolysis
- Thermal decomposition
- Catalytic converters (Zeolites and cobalt)



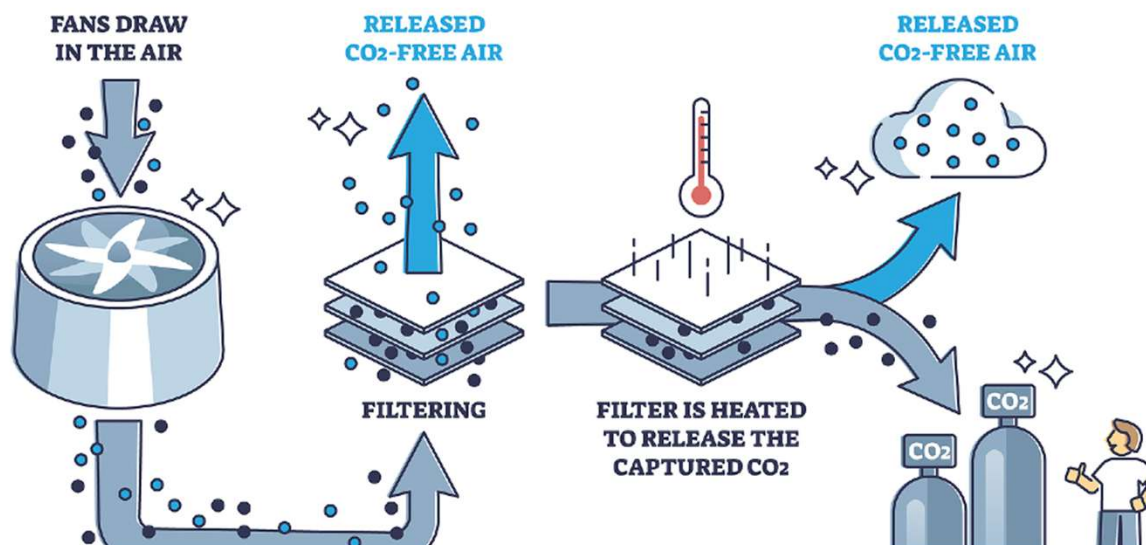
Carbon capturing technologies

- **Direct air capture and storage (DACS)** removes CO_2 directly from ambient air.
- In contrast, **carbon capture and storage (CCS)** captures CO_2 from point sources of CO_2 , such as power plants, vehicles, and factories.

CO_2 utilization contributes to a circular economy, reducing reliance on petrochemicals and mitigating climate change by converting a greenhouse gas into valuable products.

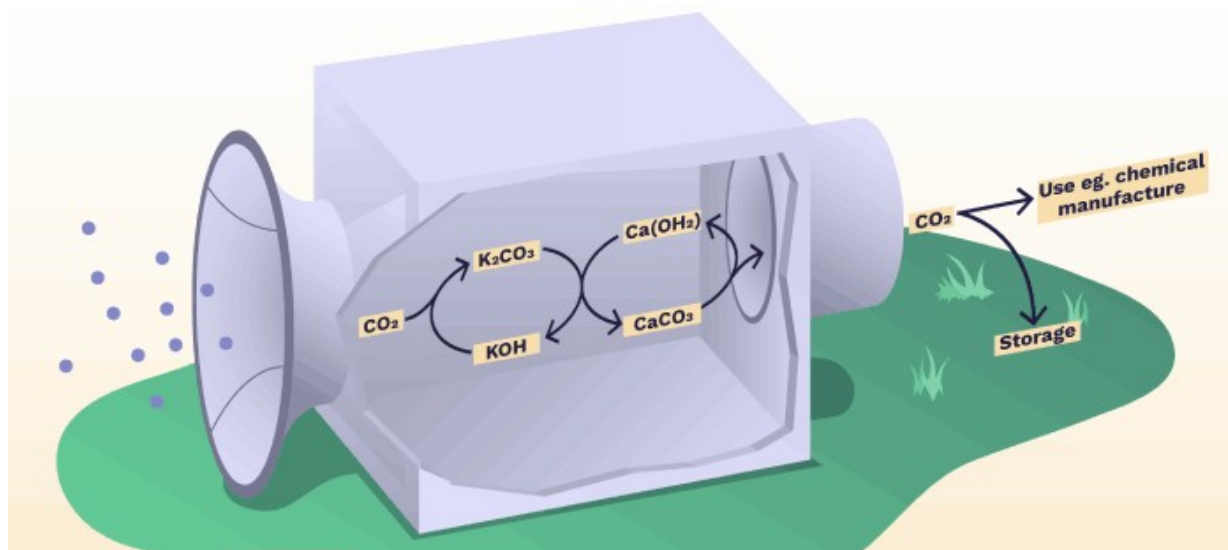


Direct air capture (DAC)



- **Direct air capture** (DAC) technology uses big machines which contains fan and filters to capture CO_2 from the air.
- When air passes through these machines, **the chemicals/materials inside the filters react with CO_2** , in order to capture it selectively and allow the rest of the air to pass through.
- Two types of capturing agents:
 1. **Liquid solvents**
 2. **Solid adsorbents**

DAC using liquid solvents



Solvent-based DAC: It requires a strong basic hydroxide solution, like NaOH or KOH to react with CO_2 , forming carbonates. This is followed by **an anionic exchange with Ca(OH)_2** that results in precipitated CaCO_3 pellets.

CO_2 must be separated from the potassium compound so that it can be used again to capture more CO_2 .

Require high temperatures ($\sim 900^\circ\text{C}$) decompose CaCO_3 to CO_2 and CaO (it reacts with water vapor giving **Ca(OH)_2**).

DAC using liquid solvents



<https://youtu.be/vuGW2InvX1A?si=YoPGS7QJQzV1y0mx>